

Windows Event Logging & Sysmon Analysis Lab

Table of Contents

1 Objective

2 Lab Environment

3 Sysmon Installation

4 Accessing Sysmon Logs

5 Event ID 1 Investigation

6 Event ID 5 Investigation

7 Event Correlation

8 Findings

9 Challenges Encountered

10 Conclusion

1. Objective

The objective of this lab was to install and configure Microsoft Sysmon, explore Windows Event Viewer, and understand how Sysmon records system activity through event logs. The lab focused on identifying process creation and process termination events and understanding their relevance in security monitoring.

2. Lab Environment

Component	Details
Operating System	Windows 11
Monitoring Tool	Sysmon
Log Viewer	Event Viewer
Shell	Windows Terminal
Scripting Environment	PowerShell
Sysmon Configuration	Default Configuration

Tools Used

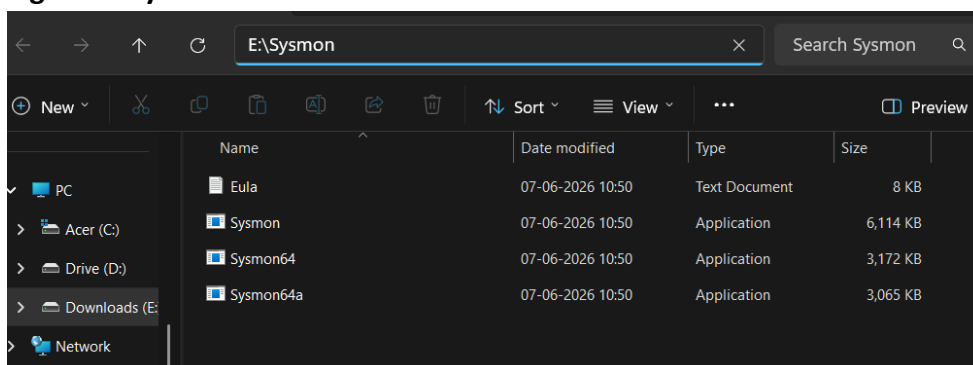
- Windows 11
 - Sysmon
 - Event Viewer
 - PowerShell
 - Windows Terminal
-

3. Sysmon Installation

Step 1: Download Sysmon

Sysmon was downloaded from Microsoft Sysinternals.

Figure 1: Sysmon Download



Observation

The Sysmon package contains Sysmon.exe and Sysmon64.exe binaries used to deploy system monitoring capabilities.

Step 2: Install Sysmon Using PowerShell

The following command was executed:

Sysmon64.exe -i

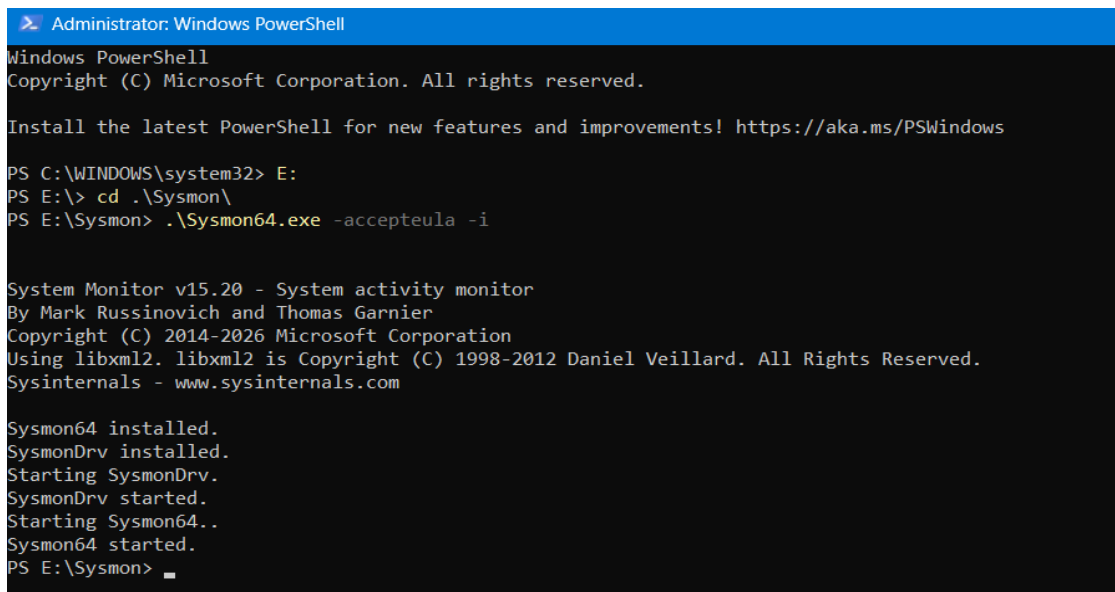
Figure 2: Sysmon Installation Through PowerShell

Observation

The installation completed successfully and the Sysmon service was registered on the system.

Analysis

PowerShell was used to execute the installation command. This activity generated process



```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\system32> E:
PS E:\> cd .\Sysmon\
PS E:\Sysmon> .\Sysmon64.exe -accepteula -i

System Monitor v15.20 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2026 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

Sysmon64 installed.
SysmonDrv installed.
Starting SysmonDrv.
SysmonDrv started.
Starting Sysmon64..
Sysmon64 started.
PS E:\Sysmon> █
```

creation events that were later recorded by Sysmon.

4. Accessing Sysmon Logs

Navigate to:

Applications and Services Logs

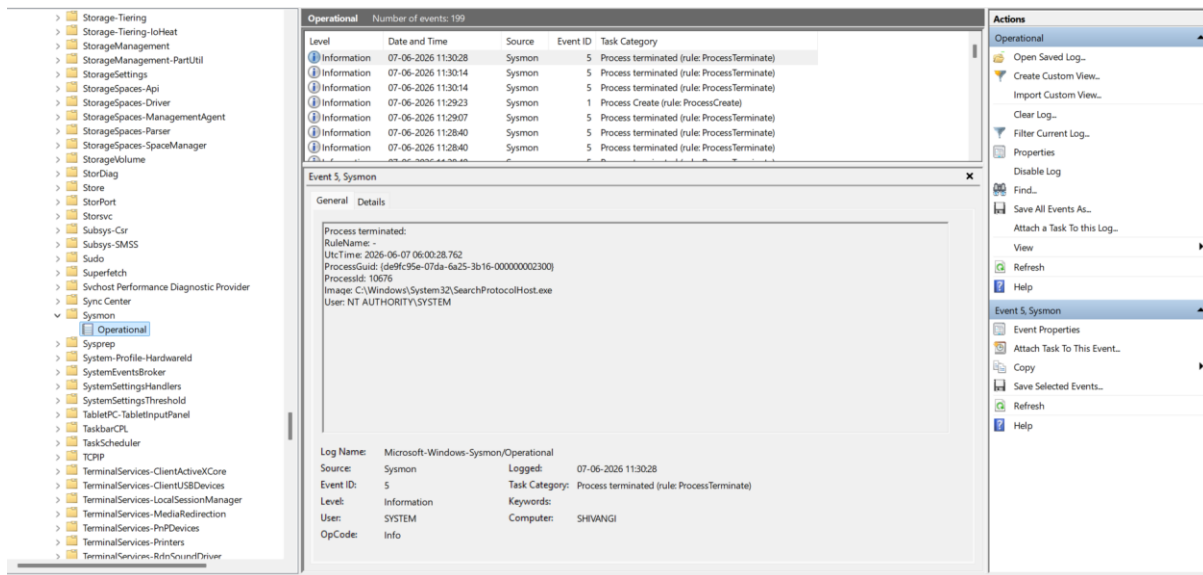
└─ Microsoft

└─ Windows

└─ Sysmon

└─ Operational

Figure 3: Sysmon Operational Log



Observation

The Sysmon Operational log contained multiple events generated after installation.

Analysis

These events provide visibility into process execution and system activity beyond standard Windows logging.

5. Event ID 1 Investigation – Process Creation

Description

Event ID 1 records process creation activity.

The event contains:

- Process Name
- Process ID
- Command Line
- User Context
- Parent Process
- Integrity Level
- Hash Values

Evidence

After Sysmon installation, the Operational log contained hundreds of Event ID 1 (Process Creation) entries generated by both Windows system processes and user activity. Due to the large number of events, manually locating the desired process was inefficient.

Evidence

Figure 4: Event ID 1 Process Creation Events

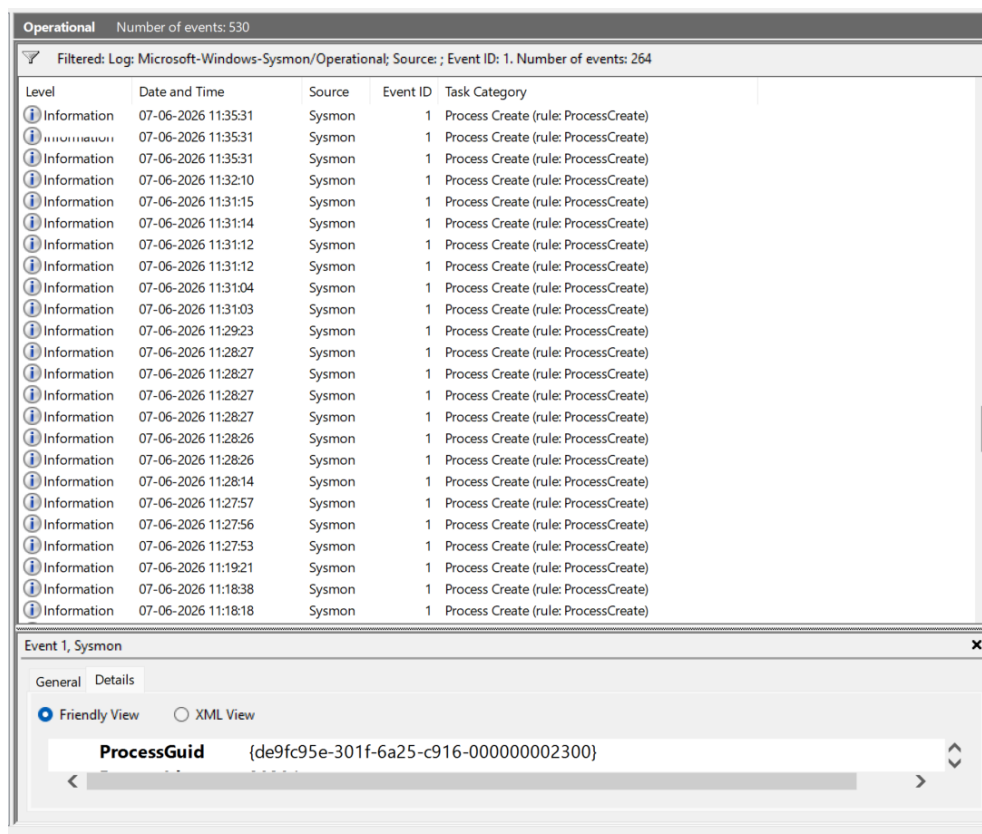
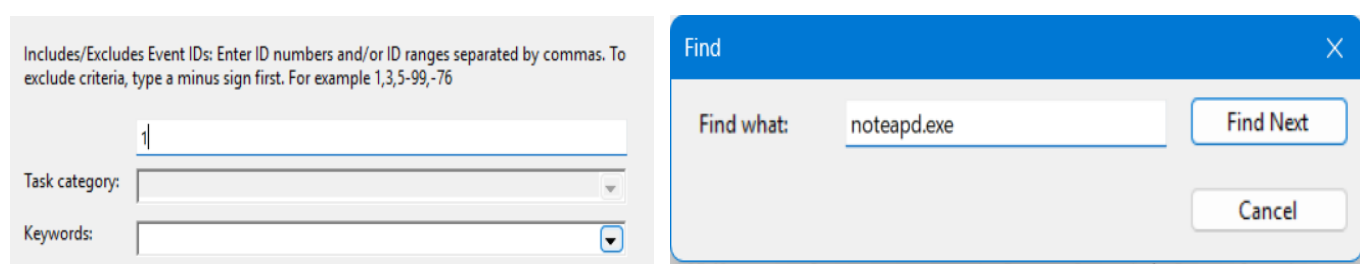


Figure 5: Event Search Using Event Viewer Find Function



Investigation Procedure

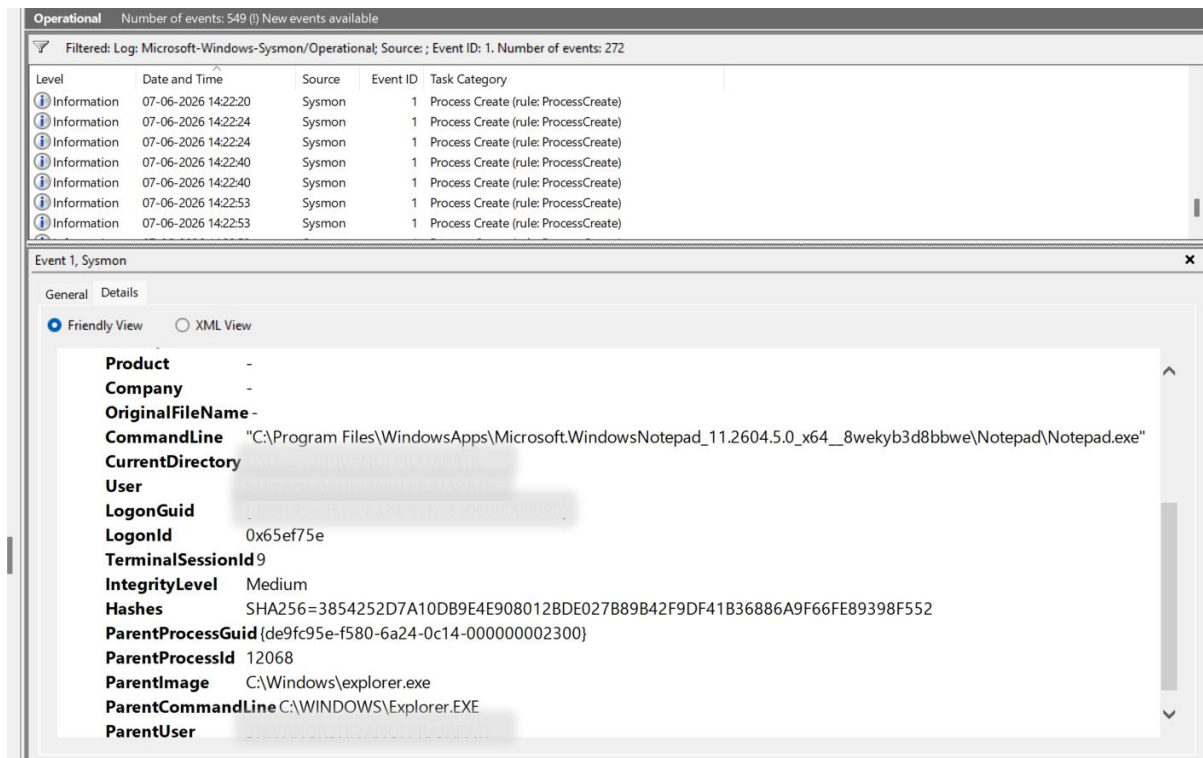
To identify the process associated with the application launched during testing, the following steps were performed:

1. Opened the Sysmon Operational log.
2. Applied a filter for **Event ID 1 (Process Creation)**.
3. Observed a large number of process creation events.
4. Used the **Find** feature within Event Viewer.
5. Searched for the keyword:

notepad.exe

6. Navigated directly to the corresponding recently generated event when Notepad was launched

Figure 6: Event ID 1 – Notepad Execution



Analysis

Notepad was launched through explorer.exe as part of normal user activity.

Parent-child relationship:

explorer.exe

└─ **notepad.exe**

This behavior demonstrates how Sysmon records user-initiated application execution.

Figure 7: Event ID 1 – Windows Terminal Process Creation

Operational Number of events: 593 (0) New events available

Filtered Log: Microsoft-Windows-Sysmon/Operational; Source: ; Event ID: 1. Number of events: 296

Level	Date and Time	Source	Event ID	Task Category
Information	07-06-2026 14:29:18	Sysmon	1	Process Create (rule: ProcessCreate)
Information	07-06-2026 14:29:19	Sysmon	1	Process Create (rule: ProcessCreate)
Information	07-06-2026 14:29:19	Sysmon	1	Process Create (rule: ProcessCreate)
Information	07-06-2026 14:29:21	Sysmon	1	Process Create (rule: ProcessCreate)
Information	07-06-2026 14:29:21	Sysmon	1	Process Create (rule: ProcessCreate)
Information	07-06-2026 14:29:21	Sysmon	1	Process Create (rule: ProcessCreate)
Information	07-06-2026 14:29:22	Sysmon	1	Process Create (rule: ProcessCreate)

Event 1, Sysmon

General Details

☒ Friendly View ☐ XML View

+ System

- EventData
 - RuleName -
 - UtcTime 2026-06-07 08:59:22.091
 - ProcessGuid {de9fc95e-32ea-6a25-2b17-000000002300}
 - ProcessId 1056
 - Image C:\Program Files\WindowsApps\Microsoft.WindowsTerminal_1.24.11321.0_x64__8wekyb3d8bbwe\WindowsTerminal.exe
 - FileVersion 1.24.2605.12001
 - Description Windows Terminal Host
 - Product Windows Terminal
 - Company Microsoft Corporation
 - OriginalFileName WindowsTerminal.exe
 - CommandLine "C:\Program Files\WindowsApps\Microsoft.WindowsTerminal_1.24.11321.0_x64__8wekyb3d8bbwe\WindowsTerminal.exe" - Embedding
 - CurrentDirectory C:\WINDOWS\system32\
 - User SHIVANGI\SHIVANGI PRAJAPATI
 - LogonGuid {de9fc95e-f57e-6a24-5ef7-5e0600000000}
 - LogonId 0x65ef75e
 - TerminalSessionId 9
 - IntegrityLevel Medium
 - Hashes SHA256=35E0F0DED9F8D1D7615D02A324E54499D17CB7C20BE5E8D599EAE8253965EB3E
 - ParentProcessGuid {de9fc95e-bf44-6a1f-1000-000000002300}
 - ParentProcessId 1352
 - ParentImage C:\Windows\System32\svchost.exe
 - ParentCommandLine C:\WINDOWS\system32\svchost.exe -k DcomLaunch -p
 - ParentUser NT AUTHORITY\SYSTEM

Observed Details

Field	Value
Process Name	WindowsTerminal.exe
User	Logged-in User
Integrity Level	Medium
Parent Process	svchost.exe
Event Type	Process Creation

Analysis

The event recorded the execution of Microsoft Windows Terminal. Sysmon captured detailed process metadata including image path, command-line information, integrity level, hashes, and parent process information.

The parent process was identified as svchost.exe, which is a legitimate Windows service host process.

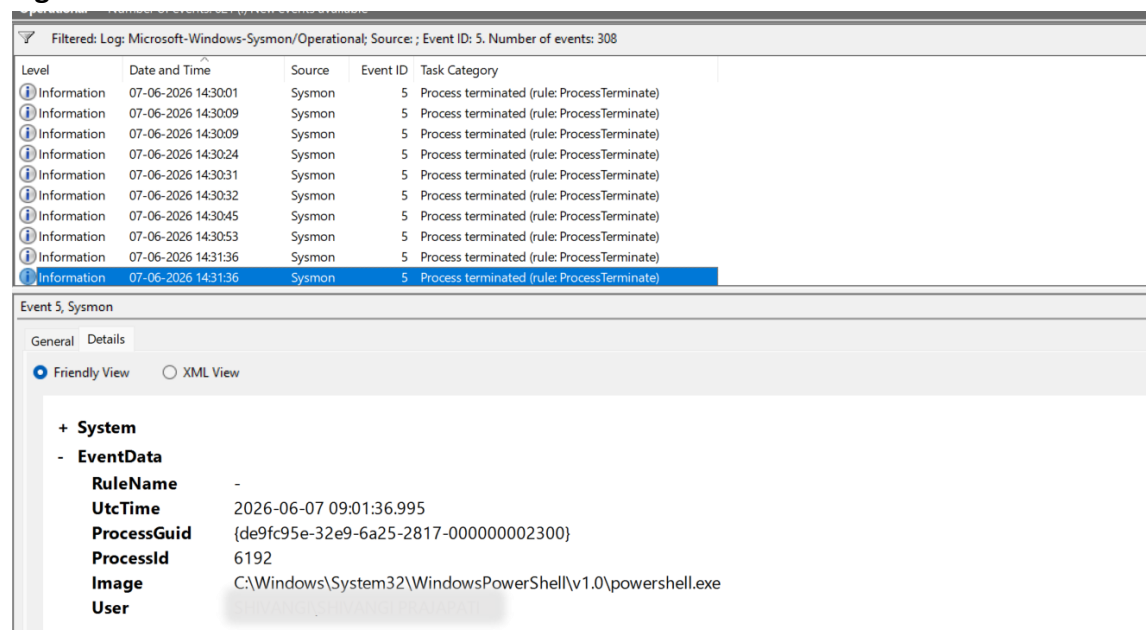
6. Event ID 5 Investigation – Process Termination

Description

Event ID 5 records process termination activity.

Evidence

Figure 8: Event ID 5 – PowerShell Process Termination



Level	Date and Time	Source	Event ID	Task Category
Information	07-06-2026 14:30:01	Sysmon	5	Process terminated (rule: ProcessTerminate)
Information	07-06-2026 14:30:09	Sysmon	5	Process terminated (rule: ProcessTerminate)
Information	07-06-2026 14:30:09	Sysmon	5	Process terminated (rule: ProcessTerminate)
Information	07-06-2026 14:30:24	Sysmon	5	Process terminated (rule: ProcessTerminate)
Information	07-06-2026 14:30:31	Sysmon	5	Process terminated (rule: ProcessTerminate)
Information	07-06-2026 14:30:32	Sysmon	5	Process terminated (rule: ProcessTerminate)
Information	07-06-2026 14:30:45	Sysmon	5	Process terminated (rule: ProcessTerminate)
Information	07-06-2026 14:30:53	Sysmon	5	Process terminated (rule: ProcessTerminate)
Information	07-06-2026 14:31:36	Sysmon	5	Process terminated (rule: ProcessTerminate)

Event 5, Sysmon	
General	
Friendly View	
+ System	
- EventData	
RuleName	-
UtcTime	2026-06-07 09:01:36.995
ProcessGuid	{de9fc95e-32e9-6a25-2817-000000002300}
ProcessId	6192
Image	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
User	

Observed Details

Field	Value
Process Name	powershell.exe
Event Type	Process Termination
User	Logged-in User

Analysis

The event confirms that the PowerShell process terminated successfully after execution. Sysmon records process termination events to provide complete visibility into process lifecycles.

7. Event Correlation

The following activities were observed during the lab:

svchost.exe

└─ WindowsTerminal.exe

└─ powershell.exe
└─ Sysmon64.exe
explorer.exe
└─ notepad.exe

Analysis

The process chains observed were consistent with legitimate administrative and user activity.

No abnormal parent-child relationships were identified.

8. Findings

1. Sysmon was successfully installed and configured.
 2. Event ID 1 successfully captured process creation events.
 3. Event ID 5 successfully captured process termination events.
 4. Windows Terminal execution was logged with detailed metadata.
 5. PowerShell activity was recorded and monitored.
 6. Notepad execution demonstrated normal user activity monitoring.
 7. Parent-child relationships were visible through Sysmon logging.
 8. No suspicious activity was identified during testing.
-

9. Challenges Encountered

- Understanding Sysmon event structure.
 - Identifying relevant Event IDs.
 - Interpreting parent-child process relationships.
 - Correlating process creation and termination events.
-

10. Conclusion

This lab demonstrated the deployment of Sysmon and the investigation of Windows process activity using Event Viewer. Event ID 1 and Event ID 5 were analyzed to understand process creation and termination behavior. Windows Terminal, PowerShell, Sysmon installation activity, and Notepad execution were successfully monitored through Sysmon logs. The exercise highlighted how Sysmon enhances endpoint visibility and provides valuable telemetry for security monitoring, threat hunting, incident response, and forensic investigations.